Kalyani Agarwal — Female, 20

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Academic Record

Qualification	Score	Institution	Year
B.Tech (Hons.) – AI & Data Science	9.39/10	Amrita Vishwa Vidyapeetham, Faridabad	2028*
Class XII (CBSE – PCM)	89.2%	St. Mary's Sr. Sec. School, Jwalapur, Haridwar	2024
Class X (CBSE)	95.8%	St. Mary's Sr. Sec. School, Jwalapur, Haridwar	2022

*Expected

Career Objective

Motivated AI & Data Science undergraduate with strong interest in Machine Learning, Deep Learning, Computer Vision, Medical Imaging AI, and intelligent software systems. Seeking internship opportunities at organisations, startups, and research labs to apply technical, analytical, and research skills in AI-driven image processing, healthcare imaging, predictive analytics, and real-world intelligent system development.

Publications & Research Experience

- **A Python-Based Investigation of Clinical Data and Ultrasound Images for PCOS Diagnosis**
  Published – STM Journal / Conference, 2025 (Completed in Semester 1 of B.Tech)
  Developed a multimodal AI-assisted diagnostic pipeline integrating ultrasound image analysis and clinical feature analytics for PCOS detection.
- **Hybrid Machine Learning Framework for Alzheimer's Disease Classification**
  Under Review
  Developed a Truncated SVD + RBF-SVM framework with statistical validation for Alzheimer's disease classification, achieving **85.81% accuracy**.
- **Autonomous Force-Aware Robotic Ultrasound for Fetal Heart Rate Monitoring**
  Under Review
  Designed a geometry-aware robotic ultrasound framework integrating inverse kinematics, trajectory planning, and convex optimisation-based force control for adaptive fetal monitoring.
- **Lightweight UAV Avalanche Monitoring Using Monocular Perception and ConvNeXtTiny with ArcFace Classification**
  Research Project
  Developed an autonomous UAV monitoring pipeline integrating monocular depth estimation, optical flow, ConvNeXtTiny-based avalanche classification, Grad-CAM explainability, and Liquid Neural Network (LNN) navigation control.
- **Explainable Multimodal AI for CFRP Structural Health Monitoring**
  Research Project
  Designed a multimodal XAI framework for CFRP strength prediction, damage segmentation, and structural integrity analysis using Hybrid Attention U-Net, MobileNetV2, SHAP, LIME, and Grad-CAM.
- **Multi-Modal Deep Learning Framework for Semiconductor Wafer Defect Analysis and Yield Prediction**
  Research Project
  Developed a multi-stream AI framework combining wafer image inspection, sensor analytics, and digital twin modelling for real-time defect detection and yield prediction.

Featured Research & Engineering Projects

SkyFare India	Developed a large-scale flight fare prediction and booking optimisation system simulating 1.14M+ domestic flights using AVL Trees, Fenwick Trees, Bitmask DP, and custom optimisation algorithms.
Attendance Analytics Dashboard	Built a full-stack attendance management platform with MySQL, Node.js, analytics dashboards, automated reporting, and role-based access control.
Delhi AQI Geospatial Analytics	Analysed 4.9L+ AQI records and developed interactive geospatial visualisations for pollution hotspot detection and temporal trend analysis.
Solar Charger with IoT Cutoff	Designed an ESP32-based smart solar charging system with automatic load cutoff, battery monitoring, and ThingSpeak cloud integration.
Smart IoT Parking System	Developed an automated IoT-enabled parking management system with slot detection, occupancy monitoring, and gate automation.
Quantum Harmonic Oscillator Simulation	Performed numerical simulation of quantum harmonic oscillator wavefunctions and computational validation of energy eigenvalue behaviour.

Technical Skills

Programming	Python, Java, HTML, CSS, JavaScript, C (basics)
Machine Learning & AI	Supervised/Unsupervised Learning, SVM (RBF), Random Forest, Logistic Regression, PCA, Truncated SVD, Transfer Learning, Explainable AI (XAI)
Deep Learning & Vision	CNNs, ResNet, MobileNetV2, Hybrid Attention U-Net, OpenCV, Medical Image Analysis, Image Preprocessing & Augmentation, Grad-CAM
Data Science & Analytics	EDA, Feature Engineering, Statistical Analysis, Model Evaluation, Time-Series Analysis, Geospatial Analytics
Mathematics & Optimisation	Linear Algebra, Probability & Statistics, Convex Optimisation, Numerical Methods, Spectral Methods
DSA & Algorithms	Data Structures & Algorithms, AVL Trees, Fenwick Trees, Hash Tables, Heaps, Bitmask DP
Robotics & Intelligent Systems	UAV Navigation, Autonomous Systems, Liquid Neural Networks (LNNs), Trajectory Planning, Inverse Kinematics
IoT & Embedded Systems	ESP32, Arduino, Sensors, MQTT, ThingSpeak
Databases & Backend	MySQL, MongoDB, REST APIs, Node.js, Express.js
Simulation & Tools	MATLAB, AirSim, Gazebo, PX4 SITL, Git, GitHub
Libraries & Frameworks	NumPy, Pandas, scikit-learn, Matplotlib

Research Domains

Artificial Intelligence & Machine Learning • Deep Learning • Computer Vision • Medical AI • Multimodal Learning • Explainable AI (XAI) • Autonomous UAV Systems • Robotics & Intelligent Control • Mathematics, Optimisation & Intelligent Systems

Achievements

- Published research in Medical AI for PCOS diagnosis using ultrasound imaging and clinical analytics (Semester 1 of B.Tech). 2025
- Developed research-oriented AI systems spanning UAVs, semiconductor intelligence, robotics, IoT, and multimodal deep learning.
- Built explainable and optimisation-driven intelligent systems involving simulation, computer vision, and autonomous control.